

Sam Shelly

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AI Engineer focused on locally-hosted, secure AI systems for privacy-sensitive environments

EDUCATION

Vassar College

Bachelor of Arts in Computer Science, Minor in Astronomy

Aug 2022 – May 2026

Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00

Relevant Coursework: Data Structures & Algorithms, Analysis of Algorithms, Software Design, Compilers, Theory of Computation, Computer Organization, Natural Language Processing, Numerical Methods, Autonomous Robotics Design

- Diagnosed recurring board failures on an Arduino Uno R4 WiFi-based robot to an undersized battery and mismatched-voltage motors frying the board, as one of two programmers on a 4-person robotics team.
- Built a written technical case – including a full engineering log – proving the failure was caused by faculty-provided component specs rather than team error; presented it to course faculty in a three-hour post-class meeting, and the analysis was accepted, correcting the grade from a C to an A-.

Danish Institute for Study Abroad

Junior Spring Semester Abroad

Jan 2025 – May 2025

Copenhagen, Denmark

Coursework: Game Development, Ethics of Artificial Intelligence

- Programmed core movement mechanics and UI for *Nightcrawler*, a 2D puzzle game built in Unity/C# with a 4-person team; published on itch.io.

EXPERIENCE

AI Development Intern

Innovative Computing Systems

May 2025 – Jan 2026

Los Angeles, CA

- Hired directly by the COO to lead a Microsoft Copilot Studio Knowledge Agent pilot at a ~60-person company; worked full-time (30 hrs/week) as the primary builder on a 4-person team, meeting weekly with the COO to scope use cases, while three engineers contributed part-time (~5 hrs/week) on team dynamics and use-case input.
- Built Python-based ingestion pipelines to index SharePoint content – including SOPs, the employee handbook, and company org hierarchy – for retrieval, and implemented an API integration connecting the agent to the company's work-ticket system, letting engineers instantly query prior work history on a given client 30% faster lookups.
- Rolled the agent out to a daily-use focus group of 15 users and began scoping it for a second prospective external client as part of early efforts to commercialize the tool.

Undergraduate Research Assistant

Vassar College

Jun 2024 – Sep 2024

Poughkeepsie, NY

- Analyzed three MAIHEM/FLASH magnetohydrodynamic simulations of a Milky Way-type galaxy's circumgalactic medium (two at solar metallicity, one at $0.3 Z_{\odot}$) to characterize turbulence-driven magnetic field evolution.
- Built visualizations using yt-project – radial Mach-Alfvén profiles, angular momentum phase plots, and magnetic/kinetic energy density evolution – showing that turbulent feedback drives disk instability and outward propagation of cold, magnetized gas that saturates the CGM's magnetic field, versus steady cubic growth in the no-feedback case.
- Co-authored and wrote roughly half of the resulting paper, presented at the KNAC Undergraduate Symposium 2024 under URSI-supported research, advised by Edward Buie II.

Computer Science Dept. Coach

Vassar College

Jan 2024 – May 2026

Poughkeepsie, NY

- Selected for a formal, paid CS Dept Coach position (8 hrs/week), retained across 4 consecutive semesters spanning two course iterations under two different instructors.
- Led 3 weekly sessions on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total), adapting pacing and session plans for struggling students.

PUBLICATIONS

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium 2024

Param P. Gogia, Umman Azan, **Sam Shelly**, Edward Buie II | *KNAC Undergraduate Symposium Proceedings*
 Investigated turbulence-driven feedback's role in magnetic field evolution in the CGM of a simulated Milky Way-type galaxy across three MAIHEM MHD simulations (two solar-metallicity, one at $0.3 Z_{\odot}$). Found that feedback induces late-stage magnetic field saturation via disk instability and outward propagation of cold, magnetized gas, versus steady cubic growth in the no-feedback case.

My contribution: performed the post-simulation analysis (Mach-Alfvén profiles, angular momentum phase plots, magnetic/kinetic energy density evolution), built all data visualizations, and wrote ~50% of the manuscript.

Efficacy of NER Models on Fiction and Nonfiction Domains 2025

Duncan Beauchamp, **Sam Shelly** | *CMPU 366 Natural Language Processing, Vassar College*
 Evaluated cross-domain performance of a DistilBERT-based NER model on literary fiction (LitBank) vs. expository text (OntoNotes), restricted to shared PER/LOC/ORG entity types. A pretrained baseline showed a 0.368 F1 domain gap between genres; fine-tuning on a mixed-domain corpus (3× oversampled LitBank, 50-token chunking, literary-only validation split) cut the gap to 0.128, raising LitBank micro-F1 from 0.26 to 0.72.

My contribution: built the training data pipeline (dataset loading, chunking, oversampling, train/validation/test splits) and wrote ~50% of the manuscript.

PROJECTS

AI Knowledge Base RAG GitHub

Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

Closed-loop RAG system built entirely on locally-hosted, open-weight components (Mistral 7B via Ollama, Qdrant) with no external API calls – suited to air-gapped and other network-restricted deployments. Applied to query government/defense AI policy documents (NIST AI RMF, OMB M-25-21, DoD AI Strategy, GSA AI compliance materials), eliminating cloud dependency and controlling data exposure for secure, privacy-sensitive environments.

- Selected PyMuPDF for PDF ingestion based on its performance reputation and ease of integration; chunks with overlap and embeds via BAAI/bge-large-en-v1.5 into Qdrant.
- Retrieves top-k chunks and generates grounded, cited answers via Mistral 7B (Ollama).
- By building a custom LLM-as-judge evaluation pipeline, achieved 0.95 retrieval recall and 0.92 answer similarity across 10 ground-truth QA pairs, visualized in a Streamlit dashboard.
- Served via FastAPI REST API, containerized with Docker Compose, with CI/CD via GitHub Actions.

DataFest 2024 — Best Overall Analysis 4-person team, 48-hr competition

Python, pandas

Analyzed student-interaction data (video views, quiz attempts, textbook page views; $n \approx 1,857$ students) from an online statistics textbook to advise an ed-tech client on improving learning outcomes.

- Team built a multiple regression model ($R^2=0.47$) identifying quiz effort and video engagement as positive predictors of student performance, and quantified that textbook-only engagement and frequent answer-changing predicted lower scores.
- A teammate built a pandas-based directed dependency graph mapping how students navigated between textbook chapters/sections while studying, informing a recommendation to prepend review material to high-traffic “capstone” chapters.
- Authored the team's final write-up and slide presentation synthesizing both analyses into client-facing recommendations; won Best Overall Analysis over 20 competing teams from 3 schools.

PPO Driving Algorithm

GitHub

Python, stable_baselines3, matplotlib, gymnasium, numpy, pygame

- Trained a 2D autonomous driving agent via Proximal Policy Optimization to navigate a gated course, achieving a 100% gate-pass rate with an asymptotic S-curve reward convergence over ~16M timesteps.
- Demonstrated policy generalization by extrapolating the trained agent to unseen course layouts achieving an 75% gate-pass rate on unseen courses.

Shader Waves

GitHub

Unity, HLSL

- Implemented real-time procedural ocean waves via direct HLSL shader programming (no Shader Graph), building the water surface from a sum-of-sines displacement model computed on the GPU.
- Derived surface normals and lighting from partial derivatives of the displacement function, working from first principles rather than using Unity's built-in tools.
- Undertaken as a graphics-programming fundamentals project, inspired by Jasper Flick's Catlike Coding series and Garrett Gunnell's water rendering work.

ACTIVITIES

Assistant Market Manager (Volunteer)

2020 – 2021

Mount Shasta Farmers Market

Mount Shasta, CA

- Opened and closed the market weekly across its ~22-week May–October season (arriving 30 min early, staying 30 min late): drove routes to place advertising signage, set up entrance, Market Match, and merchandise tents, and coordinated city-provided road-closure signage.
- Staffed an entrance checkpoint enforcing a 100-person capacity limit under COVID-19 restrictions, for a market serving 500–1,000 patrons weekly.
- Trained and directed ~3 fellow volunteers on setup and breakdown procedures.

Social Chair – Mixed Ultimate Frisbee

Aug 2022 – May 2026

Vassar College

Poughkeepsie, NY

- Served as Social Chair for the Mixed Ultimate Frisbee team, one of 3 Vassar Ultimate programs; attended weekly leadership and social-chair meetings, coordinating with the other two teams' social chairs to plan joint social events for 150+ students across all three programs.
- Planned and executed the social program for the team's annual fall home invitational, hosting 7–10 visiting colleges for a weekend tournament.
- Organized 7 consecutive nightly social events during the team's week-long spring tournament trip to Myrtle Beach, SC.

SKILLS

Languages: Python, Java, C, C++, C#, OCaml

ML/AI & Data: PyTorch, HuggingFace Transformers, DistilBERT, scikit-learn, Stable Baselines3, OpenAI Gym, LangChain, Ollama, Qdrant, pandas, yt-project

Systems & Tools: RISC-V (Computer Organization), Docker, FastAPI, Streamlit, GitHub Actions, PyMuPDF, Unity, Microsoft Copilot Studio, PyGames

Coursework/Certifications: Anthropic Academy — AI Fluency: Framework & Foundations; Claude 101

Human Languages: Spanish (Intermediate), Danish (Beginner)